Data-Driven Decision Science in the AI Era · Midterm (Week 9)

Decision Audit Midterm

Everything from W1–W4 and Ch3 in one place: how to answer, every formula, every case and number, plus a practice exam with model answers.

FORMAT	RULES	WEIGHT	TESTS
Individual Decision Audit	**Closed-book, zero coding**	**30% of the grade**	**Diagnosis, interpretation, trade-offs, recommendation**

> **The instructor's high-score standard:** *"Can you defend the costs, assumptions, risks, and trade-offs of the action?"* It is not a race to calculate fast. Every answer should end in a decision you can defend, backed by numbers, with clear limits on what you claim.

Scope: all the material you uploaded (W1, W2, W3 Ch2 extension, Ch3 labs, W4 Session 1 plus lab). The syllabus mentions Ch.1–6, but Ch5–6 are not in the material, so they are not covered here.

Study plan: Friday–Sunday

The order follows the course: the framework first, then the data, then the numbers, and a full rehearsal last. About 6 hours a day.

Friday · The framework and the data audit

- [ ] **BLOCK 1 · ABOUT 1.5 H · ANSWER TEMPLATE AND W1**
  Read the Answer template and W1. Everything else builds on this. Then write from memory: the four rules, the decision chain, and the difference between feedback, reflexivity and externality. Do Q1 and Q10.

- [ ] **BLOCK 2 · ABOUT 1.5 H · W2 BIAS–VARIANCE**
  W2: training vs test error (the U-curve), the KNN table, the Bayes classifier, model choice against a baseline. Draw the KNN table from memory. Do Q2.

- [ ] **BLOCK 3 · ABOUT 2 H · W3 DATA AUDIT**
  W3: the five audit questions, Fix/Learn/Limit, selective labels, leakage, label quality, deployment fit and the sandbox key numbers. Do Q7 and Q12.

- [ ] **EVENING · 30 MIN · CASE BANK**
  Go through the Case bank. Cover the right-hand column and say, for each case, which audit question it belongs to and which response fits.

Saturday · The numbers

- [ ] **WARM-UP · 20 MIN · RECALL FRIDAY**
  Write from memory: the five audit questions with their default responses, the four steps of the audit statement, and the K-search numbers. Check against the page.

- [ ] **BLOCK 1 · ABOUT 1.5 H · CH3 REGRESSION**
  Ch3: interpreting coefficients ("associated with"), the newspaper confounding, interaction, the four sandbox traps and the Allocation Explorer. Do Q8.

- [ ] **BLOCK 2 · ABOUT 2 H · LOGIT AND CONFUSION MATRIX**
  First half of W4: logit → probability, odds ratio, confidence intervals, the student case, every confusion-matrix metric, the accuracy trap, ROC/AUC. Calculate by hand. Do Q3 and Q5.

- [ ] **BLOCK 3 · ABOUT 2 H · COST, THRESHOLD, CALIBRATION**
  Second half of W4: break-even p*, the three applicants, the CFO table, capacity ($3,500/day) and calibration (2% → 28%). Do Q4, Q9 and Q11.

- [ ] **EVENING · 30 MIN · FORMULA SHEET**
  Write the whole formula sheet on a blank page. Check it and rewrite whatever you missed.

Sunday · Full rehearsal

- [ ] **WARM-UP · 20 MIN · FORMULA SHEET AGAIN**
  Write the formula sheet from memory once more. It should go faster than yesterday.

- [ ] **BLOCK 1 · 1 H · Q6 AS A REAL EXAM**
  Closed book, 45-minute timer. Then grade it against the model answer: does each audit point have an observation, mechanism, consequence and Fix/Learn/Limit? Does the recommendation have a scope, an owner and a stop rule?

- [ ] **BLOCK 2 · ABOUT 1.5 H · REDO THE WEAKEST**
  Pick the 3–4 practice questions where you lost the most on Friday and Saturday. Redo them without looking, and reread the section behind each mistake.

- [ ] **BLOCK 3 · 1 H · TRAPS AND CONCEPTS**
  Go through Traps and Concept cards. Say the definition and a non-example out loud before you open each card.

- [ ] **EVENING · LIGHT · FINAL PASS**
  Only the Answer template and the key numbers. No new material. Get a good night's sleep.

Answer template

The backbone of the course. Build every longer answer on this structure.

The decision chain (W1)

DATA → **MODEL** → **EVIDENCE** → **JUDGMENT** → **ACTION** → **FEEDBACK**

AI speeds up the MODEL stage. Decision science asks whether the *whole chain* is defensible. W4 extends it: Data → Probability → Threshold/Rank → Action → Human/Capacity → Outcome → Feedback.

Four rules to always include (W1)

1. **Prediction is evidence, not action.**
2. **Accuracy is not utility.**
3. **Actions can change the DGP** (data-generating process).
4. **The decision maker owns the loss function.**

The four-step audit statement (W3)

STEP	EXAMPLE
1 · Observation	The modeling file contains 0% first-time customers.
2 · Mechanism	The stated approval policy excludes that segment.
3 · Implication	The test does not evaluate that segment.
4 · Next evidence	An outcome evaluation for the proposed population.

Start with something anyone can check. Do not turn a plausible story into a fact.

The recommendation (memo and exam format)

1. **Decision and scope:** Deploy / Do not deploy / *Deploy with conditions*, and **for whom**.
2. **Evidence:** one model result (compared with a baseline) plus two data facts.
3. **Evidence gap:** what the data *cannot* tell you, i.e. who the test does not cover.
4. **Judgment:** what you claim and what you **refuse to claim**. Where does prediction end and action begin?
5. **Action:** one concrete Monday-morning action, with an **owner**.
6. **Feedback:** how the action changes future data, plus a learning plan or **stop rule**.

> A condition like "monitor carefully" does not count. Every condition needs **a metric, an owner and a trigger**. Example: *"If the 60-day default rate among pilot loans exceeds 15%, Credit Risk (owner: Head of Credit) pauses the pilot."*

Fix, Learn or Limit

RESPONSE	MEANS	TYPICAL ACTION
FIX	Change the process	Rebuild features as of the application day, redefine the label, recalibrate
LEARN	Generate better evidence	Small capped pilot, experiment, validation on a new cohort
LIMIT	Narrow the decision	Use only where evidence exists, staged rollout (5% → 20% → 50%), stop rule, rollback

AUDIT QUESTION	CORE QUESTION	DEFAULT RESPONSE
1 Representativeness	Who is in the data, and who is missing?	LIMIT + LEARN
2 Selection	How did past decisions decide what we can see?	LEARN
3 Leakage	Would we know this field at prediction time?	FIX
4 Label quality	Does the label match the outcome we care about?	FIX
5 Deployment fit	Will real use look like the way we tested?	LIMIT

The three gates: **Gate 1** Are the records credible? · **Gate 2** Is the performance claim credible (choice of K, baseline, leakage)? · **Gate 3** Does the evidence match the deployment?

W4's six questions (the pipeline template)

1. What is predicted? *(frame)*
2. What action follows? *(frame)*
3. Which error costs more? *(Move 1)*
4. What constraint limits action? *(constraint: capacity)*
5. What threshold or policy, and who owns the decision? *(Move 2)*
6. Can we trust the probability for these people? *(Move 3: calibration)*

Formula sheet

Learn these by heart. The exam is closed-book.

STATISTICAL LEARNING

$$Y = f(X) + \varepsilon$$

$$f(x) = E(Y \mid X = x) \quad \text{(regression function)}$$

$$E[(Y - \hat{f}(x))^2] = [f(x) - \hat{f}(x)]^2 + \text{Var}(\varepsilon)$$

$$\text{reducible} \qquad \text{irreducible}$$

$$E(y_0 - \hat{f}(x_0))^2 = \text{Var}(\hat{f}(x_0)) + [\text{Bias}(\hat{f}(x_0))]^2 + \text{Var}(\varepsilon)$$

$$\text{Bias} = E[\hat{f}(x_0)] - f(x_0)$$

$$\text{MSE} = \frac{1}{n} \sum (y_i - \hat{y}_i)^2$$

$$\text{Error rate} = \frac{1}{n} \sum I(y_i \neq \hat{y}_i)$$

CLASSIFIERS

$$\text{Bayes: choose the } j \text{ that maximises } P(Y = j \mid X = x_0)$$

$$\text{Bayes error} = 1 - E\left[\max_j P(Y = j \mid X) \right]$$

$$\text{KNN: } P(Y = j \mid X = x_0) = \frac{1}{K} \sum\nolimits_{i \in N_0} I(y_i = j)$$

Small K = flexible, low bias, high variance. Large K = smooth, high bias, low variance.

DECISION (W1)

$$a^*(x) = \operatorname{argmax}_a E[\, U(a, Y) \mid X = x \,]$$

The model estimates Y. The decision maker defines the actions A and the utility U.

LINEAR REGRESSION (CH3)

$$R^2 = 1 - \text{RSS}/\text{TSS}$$

$$\text{Adj } R^2 = 1 - [\text{RSS}/(n-p-1)] \,/\, [\text{TSS}/(n-1)]$$

$$\text{VIF}_j = 1 / (1 - R^2_j) \quad \text{<5 ok · 5–10 moderate · >10 serious}$$

$$P(\text{at least one false hit}) = 1 - (1 - \alpha)^P$$

$$\text{High leverage: } h > 2(p+1)/n$$

$$\text{Outlier: } |\text{studentized resid}| > 3 \cdot \text{Influential: Cook's D} > 1$$

LOGISTIC REGRESSION (W4)

$p = e^z / (1 + e^z)$, $z = \beta_0 + \beta_1 X_1 + \ldots$

$\log[p/(1-p)] = z$ (log-odds is linear)

Odds $= p/(1-p)$ · Odds ratio $= e^\beta$

z-stat $= \beta / SE$ · 95% CI $= \beta \pm 1.96 \cdot SE$

McFadden $R^2 = 1 - LL_{model} / LL_{null}$

$AIC = -2LL + 2k$ · $BIC = -2LL + k \cdot \ln(n)$

CONFUSION MATRIX

Recall/Sens = TP/(TP+FN)

Specificity = TN/(TN+FP)

Precision = TP/(TP+FP)

NPV = TN/(TN+FN)

Accuracy = (TP+TN)/N

FPR = FP/(FP+TN) = 1 − spec (Type I)

FNR = FN/(FN+TP) = 1 − recall (Type II)

Balanced acc = (recall + specificity)/2

COST → THRESHOLD (W4)

Approve: expected cost = $p \cdot C_{FN}$

Decline: expected cost = $(1 - p) \cdot C_{FP}$

Decline/flag when $p > p^* = C_{FP} / (C_{FP} + C_{FN})$

With $r = C_{FN} / C_{FP}$: $p^* = 1 / (1 + r)$

Cost per 1,000 = $(FN \cdot C_{FN} + FP \cdot C_{FP}) / N \times 1{,}000$

$r = 4 \rightarrow 0.20 \cdot r = 5 \rightarrow 0.167 \cdot r = 8 \rightarrow 0.111 \cdot r = 16 \rightarrow 0.059$

CALIBRATION AND SAMPLING (W4, OPTIONAL)

Brier = mean of $(p - y)^2$

Prior shift: $\text{logit}(p) = \text{logit}(p_{bal}) - [\text{logit}(\pi_{bal}) - \text{logit}(\pi)]$

Case-control: $\beta_0^* = \beta_0 + \log[\pi/(1-\pi)] - \log[\tilde{\pi}/(1-\tilde{\pi})]$

Uplift = P(stay | offer) − P(stay | no offer)

W1 · Prediction ≠ decision

"Model output is evidence; decisions require values, costs, actions, and accountability."

Three questions that are not the same

QUESTION	EXAMPLE	EVALUATED WITH
Prediction: what will Y be?	The customer has an 18% default risk	Test error, AUC, RMSE
Inference: how are X and Y related?	How is income related to default?	Coefficients, uncertainty
Decision: what do we do now?	At 18%: approve, reject or manual review?	Utility, loss, constraints

The same 18% can lead to very different actions depending on costs, regulation and strategy.

Basic concepts

- **Supervised** (Y exists, learn X → Y) vs **unsupervised** (only X, find structure). Whether Y exists is the first fork in the road.
- **Regression** (Y numeric) vs **classification** (Y categorical). Start from Y, not from the model name.
- **Reducible error** (better data and methods help) vs **irreducible error** (ε remains even with the true f). AI reduces some error but does not remove uncertainty.
- Flexibility vs interpretability is a **trade-off, not a ranking**. More complex does not mean more advanced.

Best prediction ≠ best action

> **The umbrella example:** 30% chance of rain. On a casual walk an umbrella is not worth it. If you carry important documents it may be. *The probability did not change; the loss function did.*
>
> **Fraud:** p = 0.20, a missed fraud costs $1,000, a false block costs $50. Let it through: 0.2 × 1,000 = **200**. Block: 0.8 × 50 = **40**. Block it. **0.5 is not a natural threshold** (here p* = 50/1,050 ≈ 0.048).

Cold open: "92% accuracy, deploy?" The right answer is C: we need the error costs and the context.

Observe ≠ intervene (Pearl)

- `P(Y | X)` is an observation: customers who received discounts repurchase more. `P(Y | do(X = x))` is an intervention: if we *give* a discount, will repurchases rise?
- Three misconceptions:
  1. Feature importance is not a policy lever. Income being important does not mean raising income reduces default.
  2. Correlation is not cause. Customers with problems simply contact support more often.
  3. Actions change the environment. After deployment, new data no longer come from the same DGP.

Feedback, reflexivity, externality

CONCEPT	DEFINITION	EXAMPLE
Feedback	System output loops back into input and changes the next state. Can be purely technical.	Recommendation → click → new training data → new recommendation
Reflexivity	People's beliefs and predictions affect actions, and the actions change the world.	Market expectations, navigation advice, credit tightening
Externality	Your action affects others without being part of your own objective.	Navigation saves you 3 minutes but moves noise and congestion into a residential area

The Moon does not change course because we predict it, but people respond to AI predictions. **Prediction → Action → Behavior → New Data.**

Guardrails

A control that stops implausible output from directly becoming action: plausible-range check, price cap, anomaly trigger, fallback or manual review (the Hsinchu case).

W2 · Complexity and generalization

"Bias = systematic blind spots; Variance = instability in decisions."

How do we estimate f?

- **Nearest-neighbor averaging:** $\hat{f}(x)$ = average of Y in the neighborhood N(x). Works well when p is small (≤ 4) and N is large.
- **Curse of dimensionality:** in many dimensions the neighbors end up far away. A 10% neighborhood is no longer local. Compare the dating app: every new filter shrinks the pool from 1,000 to 150, to 12, and finally to 1.
- **Parametric model** (linear): p+1 parameters, "almost never correct" but often a good, interpretable approximation. **Non-parametric:** assumes no form, needs much more data, and risks overfitting (Fig 2.6: zero training error).
- Trade-offs: accuracy vs interpretability, overfit vs underfit, parsimony vs black box.

Interpretability from high to low: Subset selection/Lasso → Least squares → GAM/Trees → Bagging/Boosting → SVM.

Training error vs test error

- MSE_train *always* falls as flexibility increases. MSE_test forms a **U-curve**.
- The best flexibility depends on the shape of the true f and on the noise. If the truth is nearly linear, simple models win. If it is wiggly with little noise, flexible models win.
- More flexibility means higher variance and lower bias. Choosing flexibility by test error *is* the bias–variance trade-off.

- The dartboard: low bias and low variance is the goal. High bias means tightly grouped but off target. High variance means scattered.

The KNN table (Class Activity 2)

K	FLEXIBILITY	BIAS	VARIANCE	TRAINING ERROR	GEN. RISK
1	Very high	Low	High	≈ 0	High (overfit)
10	Medium	Medium	Medium	Low–medium	Lowest (near the Bayes boundary)
100	Low	High	Low	Higher	Underfit

Fig 2.17: training error goes to 0 as 1/K → 1, while test error is U-shaped and lowest near the Bayes error rate.

Classification

- The **Bayes classifier** picks the most likely class and has the lowest possible test error (the Bayes error rate). It cannot be used in practice because the true $p_k(x)$ are unknown.
- The Covid example: n = 180, error = 20/180 = 11.1% (10 FP + 10 FN).

Model selection is a decision

> **Activity 3 (credit model):** A 92% (low stability and explainability, high monitoring effort) · B 90% (high stability and explainability, low effort) · C 91% (everything medium). There is no automatic answer. A strong answer often picks B or C and then asks: What do the errors cost? What does regulation require for explainability? How stable is it over time? *Are 2 points of accuracy worth losing explainability?*

Always compare against a **baseline** (a simple model or the existing rule). Complexity must be justified by a reliable out-of-sample improvement.

W3 · Data-to-Decision Audit

"A model can have excellent test performance and still lead to bad decisions."

Statistical generalization ≠ real-world generalization

The chain: Ch1 is the decision lens, Ch2 the model lens, W3 the data lens. The test set may differ from the people we will serve.

LEVEL	DEFINITION	THE LENDER
Target population	The people the decision concerns	All loan applicants

Sampling frame	The list or process that could include them	The application register
Observed sample	The records that reach the analysis	Approved loans only
Deployment population	The people who receive model-supported decisions	Next quarter's applicants, including first-time customers

Sampling error ≠ selection bias

- **Sampling error:** a random sample differs from its population by chance. *More rows reduce it.*
- **Selection bias:** who gets included is linked to what we want to measure. *More rows do not fix it.* "More rows do not bring back missing people."
- Note: "bias" here is about *how records were selected*, not the model bias of Ch2.

Selective labels

The decision determines which outcomes become visible. Loans: repayment is seen only if approved. Hiring: performance is seen only if hired. Bail: return to court is seen only if released.

> **The 630–639 band** (approve at ≥ 640; nobody below 640 has an outcome):
> A. Approve the whole band: creates evidence but with uncapped risk.
> B. Keep rejecting and train a more complex model: **creates no new evidence**.
> **C. A small, capped, reviewed pilot with tracked repayment: creates new evidence with bounded risk.** Best (Learn).
> D. Reject inference: *estimated, not observed*. Needs stated assumptions. This is the fallback if experiments are not allowed.

Train, validation and test inherit **the same selection process**. Random splitting does not add the missing people.

Leakage = timing

T0 application (prediction time) → T1 decision → T2 outcome (12-month default). `income_k` and `credit_score` are OK at T0. `collection_flag` is created after the outcome and must be excluded. `default_12m` is the target only. FIX: rebuild the features as they were on the application day, drop fields that did not exist then, and re-run train, validate and test. **Never pick a model by its leaked score.** 100% means leakage, not excellence.

Label quality (proxy targets)

Desired outcome (default within 12 months under the loan terms) ≠ recorded label (what the system observed) ≠ *historical approval* (an earlier decision, not an outcome). Sources of error: definition, follow-up time, recording errors, policy changes. FIX: redefine the target, or validate the proxy with a domain expert.

Deployment mismatch

EVALUATION CONDITIONS	PROPOSED USE	EVIDENCE NEEDED

Established customers	First-time customers	Coverage plus subgroup evaluation
Historical period	Next quarter	A later evaluation cohort
12 months of follow-up	Recent applications	Mature, comparable labels

"A mismatch does not prove failure. It means the earlier evidence may not be enough." LIMIT: staged rollout with checks, a stop rule and rollback.

Missing values and cleaning

- **MCAR** (random upload failure), **MAR** (the form omits income more often for new customers), **MNAR** (people with very high debt avoid reporting it). The missing percentage alone does not identify the mechanism.
- Exact duplicates are removed (same event or distinct events?). Negative income becomes missing. Large positive income is kept and investigated. Blanks get training-median imputation (who has blanks, and why?).
- Median imputation makes the software run but does not recover the true income. **dropna** removes new customers disproportionately (30% missing income vs 8% among established customers).
- "A clean report is not a deployment certificate." Technical file checks (missing values, types, balance) are different from decision-readiness questions (who is absent, what did collection measure, do labels match, were inputs available at decision time).

Audit the AI auditor (Documents A and B)

Document A says "No major issues, data quality 96/100". Document B says "DEPLOY to all next-quarter applicants: test acc 91.3% (baseline 87.5%), balanced acc 73.6%, risk low". What the AI cannot know or has not checked:

- Why first-time customers and applicants below 640 are missing (it only checks technical things)
- Whether the labels are complete (20% of defaults are recorded as 0)
- Whether every feature existed at T0
- That 96/100 is a *technical* score and "risk low" is unsupported
- The scope ("all applicants") goes far beyond the evidence
- There are no costs, no owner and no stop rule

Automated output and the reviewer's responsibility: missing-value counts → explain the recording process. Cleaning → approve the rules and check their effects. Best validation score → choose a relevant evaluation design. Predictions → judge whether the evidence supports the action.

The bank data (UCI Bank Marketing)

4,119 rows, 1,230 "unknown" cells. `duration` gives AUC 0.82 but is **leakage**, because call length is not known before you dial. `pdays = 999` means "never contacted", not 999 days. It is a schema trap that distorts distances in KNN and StandardScaler. Turn it into an indicator (previously contacted yes/no) and handle the days separately.

Ch3 · Linear regression

Advertising (200 markets), Credit, Auto. The sandbox shows four traps.

Key results

MODEL	RESULT	INTERPRETATION
sales ~ TV	β_0 7.03 · β_1 0.0475 · R^2 0.61	+$1,000 of TV is associated with about +47.5 units sold
sales ~ TV + radio + newspaper	TV 0.0458 · radio 0.1885 · news −0.0010 (p 0.86) · R^2 0.897	Newspaper is significant alone but not in the multiple model. It correlates with radio (r = 0.35) and "borrows" radio's effect.
+ TV × radio	6.75 + 0.0191 TV + 0.0289 radio + 0.0011 TV·radio · $R^2 \approx 0.97$	Synergy: the TV effect grows with the radio budget
Balance ~ Income × Student	Student +476.7 · Income:Student −2.0	Different slopes for students and non-students
Balance ~ Own / Region	No significant differences	Dummy coding: one baseline category, k−1 dummies
mpg ~ hp + hp²	R^2 0.61 → 0.69	The residual plot shows a U-pattern with the linear model, i.e. the wrong functional form

Interpreting a coefficient in multiple regression: *"Holding the other predictors fixed, a $1,000 increase in radio is associated with about 189 more units sold."* Say **associated with**, not "causes". The data are observational.

Four traps (Ch3 Sandbox)

1 · Data snooping and selection

Test P noise variables and pick the "best": the chance that at least one is significant is 1 − (1 − α)^P. With P = 40 and α = 0.05 that is **≈ 87%**. Its |β̂| is inflated even though the true β = 0. On fresh data (the second half) it is significant only about 5% of the time. Fix: select variables on one half and test on the other.

2 · R^2 inflation

R^2 can **only go up** as you add variables, even pure noise (on average k/(n−1)). Adjusted R^2 stays near 0 for noise because it charges for every extra variable.

3 · Multicollinearity

- A high correlation ρ(X1, X2) gives large standard errors. Coefficients can flip sign, and individual t-tests become insignificant while the F-test is still significant.
- $\hat{\beta}_1$ and $\hat{\beta}_2$ swing in opposite directions, but **the sum $\hat{\beta}_1 + \hat{\beta}_2$ is stable**. Predictions survive; interpretation of individual coefficients does not.
- Credit: VIF for Rating and Limit ≈ 160, for Age ≈ 1.01. Rating and Limit measure almost the same thing, so drop one or combine them.

4 · Leverage and influence

- **Leverage** means an unusual X value (h > 2(p+1)/n). An **outlier** means an unusual Y given X (|studentized residual| > 3). **Influence** means the point moves the line (Cook's D > 1).
- A high-leverage point *on* the line is harmless. With the same X *off* the line it pulls the whole slope.

Allocation Explorer: the Ch3 decision case

TV + radio = $100K. The data cover radio budgets up to $49.6K at most.

PLAN (TV/RADIO)	ADDITIVE	INTERACTION	SIMILAR MARKETS	BREAK-EVEN ($/UNIT)	PROFIT AT $30
A 100/0	—	8.66	17	52.4	−$42.7K
B 0/100	21.7	9.6	0	34.6	—
C 50/50	14.6	11.90	—	**19.4**	+$54.5K
D 70/30	—	11.26	16	22.2	+$35.4K

- **B is the trap:** the additive model says B is best (21.7), but $100K of radio lies far outside the data. It is **extrapolation** with 0 similar markets, so both models are guessing.
- Two models give two answers. The interaction model fits the data better (higher R^2).
- **The right wording:** *"C is the best listed choice under the model, a candidate plan. The data are observational, so we test the split in a few markets before committing $100K."*
- The four questions:
  1. Forecast or decide?
  2. What moves both budget and sales (confounding)?
  3. Did past decisions choose these budgets (selection)?
  4. Do we need a test?

W4 · Classification for decisions

Three moves: **Move 1** cost asymmetry · **Move 2** threshold · **Move 3** validity check (calibration).

Why logistic and not linear regression?

A straight line can give "probabilities" below 0 or above 1. Coding several classes as 1, 2 and 3 invents an order that does not exist. The S-shaped logistic curve stays in [0, 1]. The coefficient is the change in **log-odds**. The probability changes non-linearly, fastest in the middle.

The Default data (10,000 customers, 333 defaults = 3.3%)

MODEL	COEFFICIENTS	KEY POINT
balance	$-10.6513 + 0.0055 \cdot \text{balance}$ (SE 0.0002, z 25)	+$100 → +0.55 log-odds → odds ×1.73. Pseudo $R^2 = 1 - (-798.2/-1{,}460.3) = 0.45$

student	`−3.5041 + 0.4049·student`	Student 4.31%, non-student 2.92%. Students *look* riskier.
balance + income + student	`−10.869 + 0.0057 bal + 0.0030 inc − 0.6468 student`	Student −0.647 (OR 0.52). Income: CI −0.013 to 0.019 includes 0, not significant. LL −785.77, pseudo R^2 0.4619

BALANCE	Z	P(DEFAULT)
$1,000	−5.15	0.58%
$1,500	−2.40	8.3%
$2,000	0.35	58.6%

Confounding: why the sign flips

Alone: student **+0.405** (risky). With balance in the model: **−0.647** (safer).

The two-sentence answer: *Students carry higher balances on average ($988 vs $772), and balance drives default, so overall students default more (4.3% vs 2.9%). At the same balance, students default less (e.g. $1,500–2,000: 17.5% vs 25.7%), so the univariate coefficient mixes the student effect with the balance effect.*

Tell the manager NOT to reject students just for being students, and not to read −0.647 as student status *causing* lower risk.

Assumptions: log-odds linear in X, independent observations, no strong multicollinearity, no overly influential outliers, and no (quasi-)complete separation. Normal errors and homoscedasticity are *not* required. Best AIC and BIC: balance + student (1,577.7 / 1,599.3).

Confusion matrix (lender sandbox at threshold 0.50)

	REALITY: NO DEFAULT	REALITY: DEFAULT
Predict: approve	TN 2,720	FN 213 (missed default)
Predict: flag	FP 83 (false alarm)	TP 185

`Recall 46.5% · Spec 97.0% · Precision 69.0% · NPV 92.7% · Accuracy 90.8% · FPR 3.0% · FNR 53.5%`

The accuracy trap: predicting "no default" for everyone gives 2,803/3,201 = **87.6%** accuracy but finds 0 defaulters. Recall and balanced accuracy expose the problem.

Thresholds move the operating point

THRESHOLD	FLAGGED	CAUGHT OF 398	RECALL	FALSE ALARMS	PRECISION	FPR
0.50	268	185	46%	83	69%	3.0%
0.30	472	271	68%	201	57%	7.2%
0.20	624	310	78%	314	50%	11.2%
0.10	924	354	89%	570	38%	20.3%
0.05	1,245	380	95%	865	31%	30.9%

- A lower threshold catches more defaulters but declines more good applicants. A higher threshold declines fewer people but misses more defaulters. **No threshold removes the trade-off.** The overall error rate can rise while the policy improves.
- **ROC/AUC = 0.924:** the probability that a random defaulter is ranked above a random repayer (398 × 2,803 = 1,115,594 pairs, of which about 1.03 million rank correctly). One threshold gives one point on the curve. **AUC does not choose the threshold, the cost or the capacity.**
- When positives are rare: an FPR of 11% looks low, but precision is only 50%. The precision–recall baseline is the base rate (12.4%), not 50%. Report both.
- Ch4 lab (LDA on Default): threshold 0.5 → 0.2 gives TP 79 → 193, FP 22 → 232, recall 0.24 → 0.58, precision 0.78 → 0.45.

Moves 1 and 2: cost → threshold

DECISION	APPLICANT REPAYS	APPLICANT DEFAULTS
Approve	0	−8,000 (missed default)
Decline	−1,000 (false decline)	0

Baselines per 1,000 applicants: approve everyone = 398 × 8,000 / 3,201 × 1,000 = **994,689** · decline everyone = 2,803 × 1,000 / 3,201 × 1,000 = **875,664**. The model must beat both.

p* = 1,000 / 9,000 = 0.111 (lender) · **50 / 300 = 0.167** (fraud) · missed default costs 16,000 → **0.059**.

> **Three applicants** (approve = p × 8,000, decline = (1 − p) × 1,000):
>
> ```
> p=0.04: 320 vs 960 → approve
> p=0.11: 880 vs 890 → indifferent (break-even)
> p=0.30: 2,400 vs 700 → decline
> If applicant 3 really had risk 0.10: 800 vs 900 → approve. A wrong
> probability gives a wrong decision.
> ```

- Cost per 1,000 at 0.50: (213 × 8,000 + 83 × 1,000) / 3,201 × 1,000 = **558k**. At 0.10: (44 × 8,000 + 570 × 1,000) / 3,201 × 1,000 = **288k**, about half.

- The curve is **flat near its minimum** (0.07–0.11). Being roughly right about the cost is enough, but being 10 times wrong is not.
- A higher missed-default cost moves the minimum left (lower threshold). The marginal view: keep lowering the threshold as long as each extra default caught costs fewer than C_FN/C_FP false declines.
- Is the model worth it? Compare the savings against the cheapest simple policy, minus the model's running cost.

Uncertain costs (the CFO activity)

THRESHOLD	MISS 4,000	MISS 8,000	MISS 16,000	WORST EXTRA COST VS BEST
0.06	274k	301k	356k	+32% (66k)
0.10	233k	288k	398k	**+12% (≈ 42k)**
0.20	208k	318k	538k	+51% (182k)
0.33	230k	406k	758k	+113% (402k)

Choose 0.10: it stays within about 12% of the best in all three scenarios (it minimises the worst-case extra cost). **Underestimating** the missed-default cost is the more expensive mistake, because the curve is steep on the side where you decline too few. Ask for a range, not a single number.

Capacity (the fraud sandbox)

10,000 transactions a day, 5% fraud (500), missed fraud $250, false alarm $50, **capacity 500 reviews a day**.

THRESHOLD	RECALL	FALSE ALARMS	REVIEW VOLUME	DAILY COST
0.20	0.59	357	651 (over capacity)	≈ $69.1k
0.40	0.39	110	307	≈ $81.8k
0.60	0.25	36	160	≈ $95.6k
0.80	0.12	7	68	≈ $110.4k

Cost = missed × 250 + false alarms × 50. Example at 0.40: (500 − 195) × 250 + 110 × 50 = 76,250 + 5,500 = 81,750. Flag nothing: 500 × 250 = $125k.

- The break-even 0.167 flags **≈ 770 a day**, which is not feasible.
- The best feasible threshold is **≈ 0.27, about $72k a day**.
- So the capacity limit costs **≈ $3,500 a day**. Compare that with the cost of an extra reviewer.
- The threshold is not the only lever: more capacity, automated triage, or ranking ("review the top 500").

Move 3: calibration

Calibration means that among people scored about 25%, about 25 in 100 should default. Example from the sandbox: 39/152 = 25.7%, i.e. well calibrated.

> **AUC checks the order. Calibration checks the size. The break-even rule needs the size.**

GROUP	WHAT HAPPENED
Past approved loans	The points hug the diagonal
Next quarter, first-time customers	Scored 2% → **28%** defaulted. Scored 84% → 53%. Expected cost 340k, actual **1,006k** (≈ 3×)
Next quarter, established	Drifts too: 40% → 61%. Expected 321k, actual 284k
Next quarter, everyone	Predicted 33.6%, actual 33.1%. **Averages hide subgroup failures**, because the errors cancel

Responses: FIX recalibrate on recent outcomes · LEARN a small pilot with first-time customers · LIMIT use the model only for established customers. Calibration is less central for pure ranking (top 200), but the ranking and subgroup stability still need validating.

- **Measure:** reliability diagram, Brier score (p = 0.20 and no default gives 0.04; p = 0.20 and default gives 0.64), ECE. **Fix:** Platt scaling (2 parameters, works with little data) or isotonic (flexible, needs more data), fitted on data the model did not train on. *It cannot fix a population the data never covered.*
- Five models (Logistic 0.924, LDA 0.920, QDA 0.916, NB 0.910, KNN-25 0.902) are within 0.03 in AUC. Choose on Brier, stability and explainability.
- Resampling or class weights (SMOTE, 50/50) inflate the probabilities. The ranking is fine but the size is wrong. From 12.4% to 50% the shift is 1.95 on the log-odds scale. Moving the threshold keeps the calibration.
- Case-control: the slopes stay correct, only the intercept is affected. For the Default data at 50/50 the correction is ln(333/9,667) = −3.37.

From classifier to operational decision

- **Feedback changes the data:** labels are seen only for those approved, hired or reviewed. "Deployment does not only use data. Deployment creates the next data."
- **High risk ≠ persuadable** (uplift): persuadable, sure thing, lost cause, do-not-disturb. The break-even rule assumes the action works. Only a pilot or an A/B test can measure it.
- **Epic sepsis, two settings:** Michigan (38,455 hospitalizations, alert at score ≥ 6 on 18%, sensitivity 33%, PPV 12%, AUC 0.63 vs the developer's 0.76–0.83, 67% missed). Two county EDs (alerts on 4.9%, sensitivity 14.7%, PPV 7.6%, NPV 97.7%). "A vendor's threshold is a starting point. A model validated elsewhere is evidence, not permission to deploy here."
- The model P(Y = 1 | X) meets error costs, capacity, regulation, fairness, human review and business objectives. **Decision science lives in the connection between model output and action.**

Case bank

Use the cases as evidence in essay answers. One sentence on what went wrong and what it teaches is enough.

CASE	WHAT HAPPENED	LESSON AND AUDIT QUESTION
Gender Shades (2018)	Test sets were 79.6–86.2% lighter-skinned. On a balanced test: up to 34.7% error for darker-skinned women vs 0.8% for lighter-skinned men	Representativeness. An average metric hides subgroups. LIMIT
Bail decisions	Return to court is seen only for those released	Selection, selective labels. LEARN
Epic Sepsis (2021)	An input (antibiotic orders) reflects clinicians' existing suspicion. Independent AUC 0.63 vs vendor 0.76–0.83. 67% of sepsis cases missed	Leakage (timing), local validation. FIX
Optum health risk (Obermeyer 2019)	Health cost used as a proxy for health need. At the same score, Black patients were sicker. 17.7% of those flagged were Black; 46.5% if unbiased	Label quality (proxy). FIX
Zillow Offers (2021)	Price forecasts used to buy homes in a fast-changing market. $304M Q3 write-down, about 25% of staff cut, business wound down	Deployment fit. Prediction became real asset exposure: "RMSE is not just RMSE". LIMIT
SafeRent (2022–24)	Score leaned on credit history and ignored housing vouchers. $2.275M settlement	Was it tested on voucher holders? Representativeness and label
Earnest (2025)	A college's average default rate applied to the individual. $2.5M settlement	Group average in an individual score. Test for disparate impact by group
Uber (Cohen et al. 2016)	About 50M UberX observations; regression discontinuity in surge pricing to estimate price elasticity and consumer surplus	Price changes demand. The question is which objective function serves long-run value

Google Maps experiment (Arora et al. 2026)	10 U.S. cities, about 100 road segments, 6 months. Speed +2%, travel time about 0.7% better	Prediction becomes intervention and part of the traffic DGP
Cornacchia et al. (2026)	Florence, Milan, Rome: as navigation adoption rises, route diversity falls and traffic concentrates	Scale effects, externalities. Individual ETA or network welfare?
Braess's paradox	A new road can make everyone's travel time worse under selfish routing	Individual optimum ≠ system optimum. Do not label all congestion as Braess
Hsinchu rainstorm (June 2026)	A 2 km trip displayed NT$25,670–30,365. A display issue, fixed	Guardrails: range check, price cap, anomaly trigger, fallback. Not evidence about normal dynamic pricing
Dynamic pricing on a rainy night	50 people, 10 cars, normal fare NT$200	Revenue-maximising price ≠ long-run value-maximising price
UCI Bank Marketing	duration gives AUC 0.82. pdays = 999	Leakage, schema trap
Default data (ISLR)	The student coefficient flips sign	Confounding
Advertising/Allocation	The additive model recommends 100% radio, outside the data	Extrapolation, observational data ≠ effect. Test first

Key numbers from the sandbox (seed 42)

The lender: predict 12-month default at application. Historically approved: established customers with a credit score ≥ 640.

WHAT	VALUE
Register / raw loans / modeling file	`6,000 / 3,241 / 3,201 (40 duplicates)`
Train / validation / test	`1,920 / 640 / 641 (60/20/20, stratified)`
Next quarter	`3,000 applicants, 45.5% first-time customers`
Missing income / invalid (−999)	`8.9% / 32 values. New customers miss income 30% of the time vs 8%`

First-time share: register · modeling file · next quarter	`30.9% · 0% · 45.5%`
Default rate (raw file)	`12.4%`
K search: train error / val error / val bal.acc	`K=1: 0.0 / 15.3 / 0.652 · K=5: 8.5 / 11.3 / 0.691 · K=25: 9.7 / 9.1 / 0.720 · K=100: 10.5 / 10.0 / 0.617`
Test: 641 (561 no default, 80 default)	`Baseline 87.5%, 0/80, bal 50% · KNN-25 91.3%, 40/80, bal 73.6%, 16 false alarms`
Leakage (K=5)	`Bal.acc 74.7% → 100% with collection_flag`
Label miss	`20% of real defaults recorded as 0 (test: 12.5% recorded vs 15.1% oracle)`
The reveal (simulation only)	`Tested group: 15% default, bal 74% · First-time customers: 43% default, bal 58%, finds 40% (236 of about 588)`

Checking balanced accuracy: (40/80 + 545/561)/2 = (0.500 + 0.971)/2 = **0.736**. Accuracy: (40 + 545)/641 = **91.3%**.

> **Never cite oracle numbers** as evidence in a recommendation. Real lenders never have them. In the memo, use only what can be observed.

Traps that cost points

- TRAP **Choosing K (or a model) by training error.** K=1 has 0% training error and the worst validation error. Choose on validation and keep the test untouched until the end.
- TRAP **Being impressed by accuracy** without comparing with the majority baseline (87.5%). Report recall or balanced accuracy.
- TRAP **Thinking more rows fix selection bias.** They do not.
- TRAP **Removing the `segment` column** and thinking the first-time-customer problem is solved. The evidence for them is still missing.
- TRAP **Treating a missing default as repayment.** Missing means *unobserved*.
- TRAP **Using 0.5 as the default threshold.** The threshold is a result of the costs.
- TRAP **Using AUC to choose a threshold** or to show the probabilities are right. AUC only measures the order.
- TRAP **Trusting aggregate calibration.** 33.6% vs 33.1% hides the first-time customers (2% → 28%).
- TRAP **Ignoring capacity.** The break-even threshold can demand more reviews than the team can do.

- TRAP **Causal claims from observational data.** "Associated with", not "causes". Feature importance ≠ policy lever.
- TRAP **Extrapolating** beyond the range of the data (radio $100K).
- TRAP **Trusting a rising R^2** as you add variables. Use adjusted R^2 and retest on fresh data.
- TRAP **"Monitor carefully"** without a metric, an owner and a trigger.
- TRAP **Picking the most cautious label** ("pause") without naming the specific evidence. The instructor's words: "A strong answer names the specific evidence, not just the most cautious label."
- TRAP **Mixing up "bias".** Selection bias (which rows exist) is not the same as model bias (systematic error in $\hat{f}$).

Concept cards

W1's quick check: can you give a *non-example* for each term? Say the answer out loud, then open the card.

DGP –

How institutions, behavior and environment generate the data. If we only see approved loans, the sample itself is a product of the policy.

Non-example: the model itself or its hyperparameters.

Decision threshold –

A policy cut-off that turns probability into action: decline if $p \geq p^*$.

Non-example: the probability p itself.

Loss / utility –

How good or bad each action–outcome combination is for the organization: missed default −8,000, false decline −1,000.

Non-example: MSE or accuracy.

Intervention –

Actively changing a variable or policy: P(Y | do(X)). Example: raise the discount and see what happens.

Non-example: observing that discount customers buy more.

Reflexivity –

The prediction affects actions that change the system and then new beliefs. Example: navigation advice changes traffic.

Non-example: a weather forecast does not change the weather.

Feedback loop –

System output later becomes input. Example: recommendation → click → training data.

Non-example: a one-off analysis that never affects new data.

Externality –

Effects on others that are not part of your objective. Example: your shortcut brings noise to a residential street.

Non-example: your own time saved.

Guardrail –

A control that stops implausible output from becoming action: price cap, range check, manual review.

Non-example: training a better model.

Selection bias –

Who gets into the data is linked to the outcome. More rows do not help.

Non-example: sampling error in a random sample.

Leakage –

Using information that did not exist at prediction time (collection_flag, duration).

Non-example: credit score at application.

Proxy label –

A label that stands in for the real target. Example: health cost instead of health need.

Non-example: actual default within 12 months.

Calibration –

Among people scored 30%, about 30 in 100 should have the outcome. It is about size, not order.

Non-example: a high AUC.

Bias (model) –

$E[\hat{f}(x)] - f(x)$: systematic error from a model that is too simple (K=100, linear on a wiggly truth).

Non-example: predictions swinging between training sets (that is variance).

Variance (model) –

How much $\hat{f}$ changes with a different training set (K=1).

Non-example: Var(ε), the irreducible error.

Confounding –

A third variable drives both X and Y. Example: balance explains the student effect.

Non-example: an interaction effect.

Uplift –

P(outcome | action) − P(outcome | no action). Who changes behavior if we act?

Non-example: a high predicted churn risk.

Practice exam

12 questions in the instructor's style. Write your answer on paper *before* you look at the answer key after the questions. About 2 hours if you do them all.

Q1 W1 · PREDICTION VS DECISION · 8P

A churn model reaches 94% accuracy on a held-out test set. The marketing director proposes: "Send a 20% retention discount to every customer with predicted churn above 50%." Give **three** distinct reasons why the model output alone does not justify this policy. Use course concepts.

Q2 W2 + W3 · BIAS–VARIANCE · 8P

In the lender sandbox, KNN gave: K=1 train error 0.0%, validation error 15.3%; K=25 train 9.7%, validation 9.1% (best validation balanced accuracy 0.720). (a) Explain the pattern using bias and variance. (b) Your CEO asks: "Why not K = 1? It is perfect on our data." Answer in two sentences. (c) Why must the test set not be used to choose K?

Q3 W4 · CONFUSION MATRIX · 10P

A screening model is tested on 1,000 patients. TP = 60, FN = 40, FP = 90, TN = 810. (a) Compute accuracy, sensitivity, specificity, precision and balanced accuracy. (b) Compute the accuracy of "predict negative for everyone". (c) The model has lower accuracy than that rule. Is it worse? Explain.

Q4 W4 · COST, THRESHOLD, CAPACITY · 12P

An insurer flags claims for investigation. Investigating a legitimate claim costs 200; paying an undetected fraudulent claim costs 1,800. (a) Derive the break-even threshold. (b) Decide for claims with p = 0.06, 0.12 and 0.40 (show expected costs). (c) At your threshold the model flags 450 claims a day, but the team can investigate 300. What do you do, and what number would you report to management? (d) The model was trained on a 50/50 resampled dataset. Does this affect your answer?

Q5 W4 · LOGISTIC OUTPUT AND CONFOUNDING · 12P

Default data, logistic regression: intercept −10.869; balance 0.0057 (SE 0.0002); income in $1,000s 0.0030 (SE 0.0082); student −0.6468 (SE 0.2363). A model with student alone gives +0.4049. (a) Interpret the student coefficient as an odds ratio. (b) Is income significant? Show the 95% CI. (c) Explain why the student sign flips. (d) A model gives z = −2.40 for a customer. What is P(default)? (e) What should the bank manager NOT conclude?

Q6 W3 · FULL DECISION AUDIT (THE CORE QUESTION) · 20P

A hospital group trained a 30-day readmission model on patients from its urban main campus (2019–2023). Labels: "readmitted to this hospital within 30 days", from its own records. One feature is `followup_billing_code`, which is entered when a follow-up visit is billed. Test AUC is 0.85. Management wants to use the model next month at three newly acquired rural clinics to decide who gets a nurse home visit. Audit the evidence using **three** of the five audit questions (evidence, consequence, Fix/Learn/Limit) and give a recommendation with scope, owner and a stop rule.

Q7 W3 · AUDIT THE AI AUDITOR · 10P

An AI assistant drafts: "Recommendation: DEPLOY to all next-quarter applicants. Held-out test accuracy 91.3% (baseline 87.5%). Balanced accuracy 73.6%. Data quality score 96/100. Risk: low." Identify **four** problems and write two questions you would ask before signing.

Q8 CH3 · REGRESSION INTERPRETATION · 12P

Advertising data: sales ~ TV + radio + newspaper gives TV 0.0458, radio 0.1885, newspaper −0.0010 ($p = 0.86$), R^2 0.897. In a simple regression, newspaper is significant. (a) Interpret the radio coefficient. (b) Explain the newspaper result. (c) An analyst adds 15 more candidate variables, R^2 rises to 0.91 and two new variables have $p < 0.05$. Respond. (d) An additive model recommends putting the entire $100K into radio. What is wrong?

Q9 W4 · CALIBRATION · 10P

The lender's model has AUC 0.92 on past loans. Next quarter (simulation): first-time customers scored at about 2% default at 28%; established customers scored 40% default at 61%; across all applicants the average prediction is 33.6% and the actual rate is 33.1%. At threshold 0.11, model-expected cost for first-time customers is 340k per 1,000, actual cost 1,006k. (a) Why is AUC 0.92 not enough to use the threshold rule? (b) Why is the overall average misleading? (c) Choose Fix, Learn or Limit for first-time customers and justify.

Q10 W1 · FEEDBACK AND REFLEXIVITY · 8P

A navigation app recommends the fastest route to millions of drivers. (a) Distinguish feedback, reflexivity and externality using this example. (b) What does the Google Maps routing experiment show for decision science? (c) What objective question must the platform answer?

Q11 W4 · COST UNCERTAINTY · 8P

The CFO says a missed default costs between 4,000 and 16,000 (best guess 8,000); a false decline costs 1,000. Total cost per 1,000 applicants: threshold 0.06: 274k / 301k / 356k; 0.10: 233k / 288k / 398k; 0.20: 208k / 318k / 538k; 0.33: 230k / 406k / 758k (for miss cost 4k / 8k / 16k). Commit to one threshold and defend it.

Q12 W3 · SELECTIVE LABELS · 8P

The lender approves only applicants scoring 640 or above. Nobody below 640 has a repayment record. The CEO wants to know whether applicants in the 630–639 band are creditworthy. Options: A approve everyone in the band; B keep rejecting and train a more complex model; C approve a small, capped, reviewed sample and track repayment; D estimate their outcomes from similar accepted borrowers (reject inference). Which options create new evidence? Recommend one, and say what you would do if no experiment is allowed.

Answer key

Model answers for Q1–Q12. Grade yourself after writing your own answer.

Q1 · W1 · PREDICTION VS DECISION · 8P

1. **Accuracy is not utility, and it may hide a weak model.** If only 6% churn, predicting "no churn" for everyone already gives 94%. We need recall, precision or balanced accuracy compared with that baseline.
2. **The 0.5 threshold is not derived from costs.** The discount cost (C_FP: giving it to someone who would stay) and the lost customer value (C_FN) set the break-even p* = C_FP/(C_FP + C_FN). If a lost customer is worth far more than the discount, p* is well below 0.5.
3. **High risk is not the same as persuadable.** The model estimates P(churn | X), not the effect of the offer, P(stay | offer) − P(stay | no offer). Some high-risk customers leave anyway; some stay anyway. Only an A/B test or pilot measures uplift.
4. (Alternatives) **Feedback/DGP change:** discounts change behaviour and future training data. Customers may learn to threaten churn. **Calibration:** the policy uses the size of p, which needs checking per segment. **Capacity/budget** constraints.

Scoring tip: three *different* reasons, each tied to a course concept. End with what you would do instead (a pilot with a control group, a threshold derived from costs).

Q2 · W2 + W3 · BIAS–VARIANCE · 8P

(a) Small K is very flexible: low bias, high variance. K=1 memorises each training point (0% training error) including noise, so it generalises poorly (15.3%). Larger K averages over more neighbours, adding some bias but cutting variance. K=25 sits near the bottom of the U-shaped validation curve. K=100 starts to underfit (validation error 10.0%, balanced accuracy 0.617).

(b) "K = 1 fits the past perfectly because it memorises it, noise included; on new applicants its error is the highest of all options (15.3%). We chose K = 25 because it had the best validation performance, and new applicants are what the decision is about."

(c) The test set gives one untouched, final check of the chosen model. If we use it to choose K, we optimise on it and its score becomes optimistically biased; we no longer have an honest estimate. Validation chooses, test confirms once.

Bonus: all three parts (train/validation/test) still inherit the same selection (approved, established customers only). Good generalization here does not prove the model works for first-time customers.

Q3 · W4 · CONFUSION MATRIX · 10P

```
Accuracy = (60+810)/1000 = 87.0%
Sensitivity = 60/100 = 60.0%
Specificity = 810/900 = 90.0%
Precision = 60/150 = 40.0%
Balanced acc = (0.60+0.90)/2 = 75.0%
All-negative accuracy = 900/1000 = 90.0% (sensitivity 0%, balanced 50%)
```

(c) Not necessarily. The all-negative rule misses every one of the 100 sick patients. The model finds 60 of them at the cost of 90 false alarms. Which is better depends on the costs: if a missed disease is far more costly than an extra test, the model is much better even though accuracy is lower. Accuracy treats all errors as equal; the decision does not.

A precision of 40% means 6 in 10 flagged patients are healthy. Mention it when you discuss the cost of following up the flagged cases.

Q4 · W4 · COST, THRESHOLD, CAPACITY · 12P

(a) Investigate when $p \times 1{,}800 > (1 - p) \times 200 \rightarrow p > 200/(200 + 1{,}800) =$ **0.10**.

```
(b) p=0.06: pay 108 vs investigate 188 → pay
p=0.12: pay 216 vs investigate 176 → investigate
p=0.40: pay 720 vs investigate 120 → investigate
```

(c) Capacity binds. Rank claims by p and investigate the top 300 (equivalently, raise the threshold until about 300 are flagged). Report the **cost of the capacity limit**: the expected cost at the feasible threshold minus the cost at the break-even threshold. Management can compare it with the cost of extra investigators or automated triage. The threshold is not the only lever.

(d) Yes. Resampling to 50/50 inflates predicted probabilities (ranking/AUC may be fine, size is wrong), so comparing p with 0.10 would flag too many claims. Recalibrate first (shift the log-odds back to the true base rate, or Platt/isotonic on untouched recent data), or set the threshold on ranks. Check calibration by claim type before deploying.

Q5 · W4 · LOGISTIC OUTPUT AND CONFOUNDING · 12P

(a) e^−0.6468 ≈ **0.52**: holding balance and income fixed, a student has about half the odds of default of a non-student. z = −0.6468/0.2363 = −2.74, p = 0.006; CI −1.110 to −0.184 excludes 0.

(b) 0.0030 ± 1.96 × 0.0082 = **−0.013 to 0.019**, which includes 0 (z = 0.37, p = 0.71). Not significant.

(c) Confounding by balance: students carry higher balances on average ($988 vs $772), and balance drives default. Alone, student status picks up that balance effect, so students look riskier (4.3% vs 2.9%). At the same balance students default less (e.g. $1,500–2,000: 17.5% vs 25.7%).

(d) p = e^−2.40/(1 + e^−2.40) = 0.0907/1.0907 ≈ **0.083 (8.3%)**.

(e) Not that student status causes lower risk, and not to reject students as a group based on the univariate result. The coefficient is an association, conditional on the other variables. The decision should use the full risk estimate plus costs, not one feature.

Q6 · W3 · FULL DECISION AUDIT (THE CORE QUESTION) · 20P

Leakage (Fix). `followup_billing_code` is created after discharge, often after the event we predict. It would not exist at the decision time (discharge). The 0.85 AUC may be inflated. Fix: rebuild features as they existed at discharge, drop the field, re-run train/validation/test.

Label quality (Fix). The label counts only readmissions to *this* hospital. Patients readmitted elsewhere are recorded as "not readmitted". This is likely more common for rural patients far from the main campus, so the label undercounts exactly where we will deploy. Fix: link regional or insurer data, or validate the proxy with clinicians.

Representativeness / deployment fit (Limit + Learn). Rural clinic patients are 0% of the training data; distance, access and case mix differ. 2019–2023 includes the pandemic period. The test evaluates urban patients only. Limit: do not use it for allocation at rural clinics yet; Learn: run the model in shadow mode (scores recorded, not used) and collect outcomes.

Recommendation: Deploy with conditions. After removing the leaked field and re-validating, use it at the main campus only. At rural clinics, run a 3-month shadow pilot: home-visit decisions follow the current process while scores are logged. **Owner:** the Director of Care Coordination. **Stop rule / trigger:** before any rural use, calibration by clinic must be within ±5 percentage points of observed readmissions; if not, recalibrate or keep the current process.

Claim / refuse to claim: I claim the model ranks urban patients reasonably well once the leak is removed. I do not claim it works for rural patients, or that a home visit reduces readmission. That needs an uplift test.

Each audit point needs an observation, mechanism, consequence and response. Selection is an alternative fourth choice: we only see admitted patients.

Q7 · W3 · AUDIT THE AI AUDITOR · 10P

1. **Scope exceeds evidence:** the test set is a slice of approved, established customers. First-time customers are 0% of the file but 45.5% of next quarter.
2. **Weak improvement:** +3.8 points over a rule that learns nothing, and the model still misses 40 of 80 defaults (recall 50%).
3. **Data quality ≠ decision readiness:** 96/100 checks duplicates, ranges and missing values, not who is absent, label completeness or feature timing.
4. **"Risk: low" is unsupported:** no costs, no threshold, no calibration check, no subgroup evaluation, no owner or stop rule. Selective labels mean rejected applicants' outcomes are unknown.

Questions: "How does the model perform for first-time customers, and where is that evidence?" · "Was every feature available at application time, and how complete are the default labels?" (Also good: "What do the errors cost, and what threshold follows?")

Q8 · CH3 · REGRESSION INTERPRETATION · 12P

(a) Holding TV and newspaper fixed, an extra $1,000 on radio is *associated with* about 189 more units sold. It is an observational association, not a proven effect.

(b) Newspaper spending is correlated with radio ($r \approx 0.35$). In the simple regression newspaper picks up radio's effect; once radio is in the model, newspaper adds nothing. This is confounding/omitted-variable bias.

(c) R^2 can only increase when variables are added, even pure noise, so the rise means little: check adjusted R^2. With 15 tests at $\alpha = 0.05$, the chance of at least one false positive is $1 - 0.95^{15} \approx 54\%$, so two "significant" variables are expected by chance. Validate on fresh data (select on one half, test on the other).

(d) Extrapolation: the largest radio budget in the data is $49.6K, and no similar market exists for $100K radio. The additive model also ignores the TV×radio synergy (the interaction model fits better, $R^2 \approx 0.97$). The best listed in-range plan was 50/50, a candidate to *test* in a few markets before committing $100K, because the data are observational.

Q9 · W4 · CALIBRATION · 10P

(a) AUC checks only the order of the scores. The break-even rule compares the size of p with p* = 0.11, so it needs calibrated probabilities. A first-time applicant scored 2% is approved although the true risk (28%) is far above 0.11, so the policy approves many bad loans and the actual cost is about three times what the model expects.

(b) Errors in different groups cancel: first-time customers are under-predicted at low scores and over-predicted at high scores (84% → 53%), while established customers are under-predicted. The aggregate looks perfect while subgroups fail. Calibration must be checked for each group we serve.

(c) Limit now, plus Learn: use the model only for established customers (where it has evidence) and keep first-time customers on the current process, while a small capped pilot collects their outcomes. Then Fix: recalibrate or retrain once mature labels exist. Recalibration alone cannot fix a population the training data never covered.

Q10 · W1 · FEEDBACK AND REFLEXIVITY · 8P

(a) **Feedback:** recommendations shift traffic, the new traffic data feed the next prediction. **Reflexivity:** drivers respond to the prediction; if 80% switch to "faster" route B, B is no longer fastest. **Externality:** one driver saves minutes while noise and congestion move to a residential street, a cost not in the driver's or app's objective.

(b) Arora et al. (2026) adjusted route costs on about 100 congested segments in 10 U.S. cities for six months: speed rose about 2% and travel time improved about 0.7%. A routing app is an intervention that changes the traffic DGP, not just a predictor. Cornacchia et al. (2026) show that at high adoption, routes concentrate and benefits can weaken or reverse.

(c) Optimise individual ETA or network welfare? Individual optimisation ≠ system optimisation (Braess's paradox is one specific case of this; do not label all congestion as Braess).

Q11 · W4 · COST UNCERTAINTY · 8P

Best possible per scenario: 208k (4k), 288k (8k), 356k (16k). Extra cost (regret) of each threshold in its worst case: 0.06 → +66k (+32%); **0.10 → +42k (+12%)**; 0.20 → +182k (+51%); 0.33 → +402k (+113%).

I choose 0.10 because it stays within about 12% of the best achievable cost whatever the true miss cost is, and it is optimal at the best guess (288k). It matches the break-even for 8,000 (p* = 0.111). 0.20 looks best if misses are cheap, but if they really cost 16,000 we pay 51% more. Underestimating the miss cost is the more expensive mistake, because the cost curve is steep on the side where we decline too few. I would ask Finance to narrow the range and revisit the threshold quarterly.

Q12 · W3 · SELECTIVE LABELS · 8P

New *observed* evidence comes only from A and C. B creates none: a more complex model trained on the same selected file still has no outcomes below 640. D produces estimates, not observations, and rests on the assumption that rejected applicants behave like similar accepted ones.

Recommend C (Learn): a small, capped pilot (e.g. 100 loans, a maximum exposure set by Credit Risk) with manual review and 12-month tracking. It creates evidence with bounded risk. Owner: Head of Credit Risk. Stop rule: pause if early delinquency in the pilot exceeds a preset level. Requires legal and compliance review.

If no experiment is allowed: use D with the assumptions stated openly, and Limit: keep the band on the current process. Report the result as an estimate, not evidence.

> **Grade yourself:** does every answer have a decision, a number, a course concept and a limit on what you claim? If yes, you are at the high-score level.

Built from the course material you uploaded (W1–W4, Ch3, sandboxes and labs). Simulation numbers refer to seed 42 (the default run).